

Umair Yousaf

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Education

University of Colorado Boulder, MS in Computer Science - CGPA 4.0 Aug 2025 – May 2027

- **Courses:** Distributed Systems, Intro. to Robotics, Transformers for Robotics, Neuro-Symbolic NLP, Linear Programming

Lahore University of Management Sciences (LUMS), BS in Computer Science - CGPA 3.6 Aug 2019 – May 2023

- **Courses:** NLP, Deep Learning, Computer Vision, ML, Data Mining, Statistics & Data Analysis, Applied Probability

Technologies

Robotics: ROS2, NVBlox, Open3D, PCL, RViz2, RTDE, UR5e, Franka Pandalas, Clearpath Ridgeback, RGB-D Perception

ML / AI: PyTorch, TensorFlow, HuggingFace, LangChain, LangGraph, CUDA, cuML, YOLO, Vision Transformers, RAG

Cloud & Tools: Azure (ML, Container Instances, DevOps), Docker, Kafka, Spark, gRPC, Flask, Git

Experience

NASA / CU Boulder Collaborative Robotics Project, Graduate Researcher - Boulder, CO May 2026 – Present

- Developing force-constrained motion primitives for collaborative robot manipulation with guaranteed force bounds, hosted at **PickNik Robotics** - target application: crew-safe robotic assistance on lunar space stations.
- Implementing safety-guaranteed control strategies in **ROS2/MoveIt Pro** on a dual-arm platform (**2x UR5e arms on Clearpath Ridgeback**), targeting safe physical human-robot interaction under contact dynamics.

NSF Institute for Student-AI Teaming, Graduate Researcher - Boulder, CO Feb 2026 – Present

- Developing an AI-assisted classroom discourse monitoring system for student collaborative learning, applying **neuro-symbolic NLP** techniques with **LLaMA 3.3 70B** for real-time utterance classification and dialogue management.
- Built a similarity-based dialogue router using **cosine embeddings** to classify and route student utterances, reducing LLM misclassification; migrated vector store from ChromaDB to **Qdrant** for improved retrieval performance.
- Integrated LLMs with real-time speech detection using **Whisper** and **SileroVAD** for noisy classrooms; system deployed and evaluated across live classroom pilots in K-12 settings.

Reteta, Junior Consultant (AI Team) - Cranbury, NJ · Remote Nov 2024 – Jul 2025

- Fine-tuned LLMs on patient-doctor interaction data to generate structured SOAP notes with **95% accuracy**, reducing physician documentation time by **75%**.
- Deployed and maintained a scalable ML backend on **Azure** using Azure ML and Azure Container Instances; built CI/CD pipelines with **Azure DevOps** for automated testing, deployment, and monitoring.
- Developed **NER-based PII anonymization** pipeline ensuring HIPAA compliance across all patient data workflows.

Visionet Systems, Junior Consultant (AI Team) - Lahore, Pakistan Jul 2023 – Nov 2024

- Developed a PDF parsing pipeline integrated with **Agentic AI and RAG** architectures, improving document retrieval efficiency by **20%**; pipeline extracted text, tables, images, captions, headers, and footers across large insurance corpora.
- Built end-to-end Generative AI solutions for insurance underwriting and claim management workflows using **LangChain** and **LangGraph**, streamlining client operations.
- Built a clothing recommendation and reverse image search system using transformers, Siamese networks, and contrastive learning, improving recommendation accuracy by **15%**.
- Trained multimodal LLMs for VQA and image-based extraction.

Aletheia-AI (now BrickAndMortar.AI), ML & SWE Intern - Lahore, Pakistan Jun 2022 – Aug 2022

- Deployed real-time detection systems using **YOLOv7** for vehicle, ANPR, and person detection, achieving **>90% detection accuracy** in production.

Projects

Real-Time Tabletop Perception and Manipulation Pipeline | Robotics - CU Boulder

- Built a real-time **RGB-D** perception pipeline (20 FPS) for tabletop object detection and 7-DOF pose estimation.
- Integrated volumetric fusion and ESDF computation using **NVBlox** for collision-aware motion planning; implemented table-plane removal, robot geometry filtering, and GPU-accelerated **DBSCAN** clustering via cuML.

LEGAL-UQA: Low-Resource Urdu-English Legal QA Dataset | NLP/ML | [arXiv preprint](#)

- Created the first Urdu-English legal QA dataset (619 Q&A pairs) from Pakistan's constitution using OCR extraction, manual refinement, and **GPT-4**-assisted translation.
- Evaluated generalist LLMs and embedding models; achieved **99.19% human-evaluated accuracy** with Claude-3.5-Sonnet. Fine-tuned **mt5-large-UQA-1.0** for domain adaptation; benchmarked retrieval showing OpenAI text-embedding-3-large outperforms Mistral embeddings.